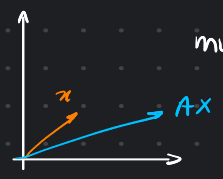
॥श्री गणेशाय नमः॥

Projections and Transformations

Linear transformations



multiplying a matrix to a vector either rotates or scales the vector.
↳ transforms x to another vector.

Properties of a linear structure:

① additivity: $T(a+b) = T(a)+T(b)$

② homogeneity $T(aX) = aT(X)$

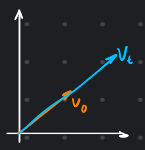
Transformation: $T: V \to W$ (V: original vector space, W: transformed space)

e.g. $T: R \to R$, $T = x^2$

$x_0 = 2 \Rightarrow x_1 = 4$

Transformation applied on any input (scalar/vector/matrix) in one vector space will result in output (scalar/vector/matrix) in another (or same) vector space

$T(x_1+x_2) = (x_1+x_2)^2 \neq x_1^2 + x_2^2$ NOT linear.



$T(V_0) = V_1$

Is it linear?

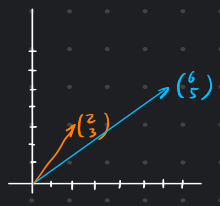
→ coeff on y-axis = 0 before transformation
= 0 after transformation

linear

$T = (V_1, V_2, V_1+V_2)$, $T: \mathbb{R}^2 \to \mathbb{R}^2$

$V_0 = \begin{pmatrix}2\\3\end{pmatrix}$

$T(V_0) = \begin{pmatrix}2+3\\2+3\end{pmatrix} = \begin{pmatrix}6\\5\end{pmatrix} = V_1$



This does both rotation & scaling

Is T linear?

$T(V_1+V_2) \stackrel{?}{=} T(V_1)+T(V_2)$

$T\begin{pmatrix}a_1+b_1\\a_2+b_2\end{pmatrix} \stackrel{?}{=} T\begin{pmatrix}a_1\\a_2\end{pmatrix} + T\begin{pmatrix}b_1\\b_2\end{pmatrix}$

$\begin{pmatrix}a_1a_2+a_1b_2+a_2b_1+b_1b_2\\a_1+a_2+b_1+b_2\end{pmatrix} \stackrel{?}{=} \begin{pmatrix}a_1a_2\\a_1+a_2\end{pmatrix} + \begin{pmatrix}b_1b_2\\b_1+b_2\end{pmatrix}$

$\neq$

NOT linear

$T: \mathbb{R}^2 \to \mathbb{R}^2$

a) $T = (V_1+V_2, V_1-V_2)$

$T\begin{pmatrix}a_1+b_1\\a_2+b_2\end{pmatrix} = \begin{pmatrix}a_1+b_1+a_2+b_2\\a_1-a_2+b_1-b_2\end{pmatrix}$

$= \begin{pmatrix}a_1+a_2\\a_1-a_2\end{pmatrix} + \begin{pmatrix}b_1+b_2\\b_1-b_2\end{pmatrix}$

$= T\begin{pmatrix}a_1\\a_2\end{pmatrix} + T\begin{pmatrix}b_1\\b_2\end{pmatrix}$

b) $T = (0, V_1+V_2)$

$T\begin{pmatrix}a_1+b_1\\a_2+b_2\end{pmatrix} = \begin{pmatrix}0\\a_1+a_2+b_1+b_2\end{pmatrix}$

$= \begin{pmatrix}0\\a_1+a_2\end{pmatrix} + \begin{pmatrix}0\\b_1+b_2\end{pmatrix}$

linear

c) $T = (1, V_2-V_1)$

$T\begin{pmatrix}a_1\\a_2\end{pmatrix} + T\begin{pmatrix}b_1\\b_2\end{pmatrix} = \begin{pmatrix}1\\a_2-a_1\end{pmatrix} + \begin{pmatrix}1\\b_2-b_1\end{pmatrix}$

$= \begin{pmatrix}2\\a_2+b_2-a_1-b_1\end{pmatrix}$

non-linear

d) $T = (1,1)$ ← any scalar would make T non-linear

$\begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}1\\1\end{pmatrix} \neq \begin{pmatrix}1\\1\end{pmatrix}$

not linear

e) $T = (0,0)$

$\begin{pmatrix}0\\0\end{pmatrix} + \begin{pmatrix}0\\0\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$

linear

f) $T = (V_1^2+V_2^2, V_1^2-V_2^2)$ ← non-linear components

non-linear

g) $T = (|V_1|, V_2)$

non-linear

h) $T = \left(\frac{V_1}{V_2}, V_1\right)$

non-linear

i) $T=(v_2, v_1)$

$\begin{pmatrix}a\\b\end{pmatrix}\ \begin{pmatrix}c\\d\end{pmatrix}$

$T\begin{pmatrix}a+c\\b+d\end{pmatrix}=\begin{pmatrix}b+d\\a+c\end{pmatrix}$

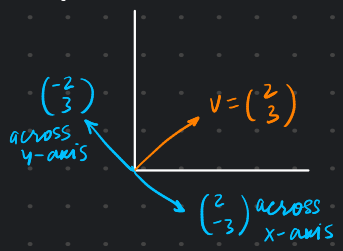
$T\begin{pmatrix}a\\b\end{pmatrix}+T\begin{pmatrix}c\\d\end{pmatrix}=\begin{pmatrix}b\\a\end{pmatrix}+\begin{pmatrix}d\\c\end{pmatrix}=\begin{pmatrix}b+d\\a+c\end{pmatrix}$

linear.

Other linear transformations (apart from scaling and rotaⁿ)

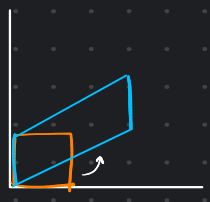
Reflection

(does rotaⁿ)



Shearing

(rotaⁿ + scaling)



more examples:

$T: V \to W$

$\begin{pmatrix}x_1\\\vdots\\x_m\end{pmatrix} \to \begin{pmatrix}x_1\\x_2\\\vdots\\x_m\end{pmatrix}$

a) $T: \mathbb{R}^2 \to \mathbb{R}^3$

$T=(x_1, x_2, 3)$

non-linear

b) $T: \mathbb{R}^3 \to \mathbb{R}^2$

$T=(x_1, 0)$

linear

c) $T: \mathbb{R}^2 \to \mathbb{R}^3$

$T=(x_1, x_2, 0)$

linear

d) $T: \mathbb{R}^3 \to \mathbb{R}$

$T=(c)$ c is a scalar

non-linear

e) $T: \mathbb{R}^2 \to \mathbb{R}^3$

$T=(0, x_2, x_1)$

linear

f) $\mathbb{R}^2 \to \mathbb{R}^3$

$T=(x_1, x_2, x_2, x_1)$

non-linear

g) $T: \mathbb{R}^2 \to \mathbb{R}^2$

$T=(\sin x_1, x_2)$

non-linear

$T=Av$ A is a matrix

$T: \mathbb{R}^3 \to \mathbb{R}^3$

Is it linear?

$T\begin{pmatrix}a\\b\end{pmatrix}+T\begin{pmatrix}c\\d\end{pmatrix}=A\begin{pmatrix}a\\b\end{pmatrix}+A\begin{pmatrix}c\\d\end{pmatrix}=A\begin{pmatrix}a+c\\b+d\end{pmatrix}=T\left[\begin{pmatrix}a\\b\end{pmatrix}+\begin{pmatrix}c\\d\end{pmatrix}\right]$

linear

General form of linear transformations

$T(v)=Av$ (matrix multiplicaⁿ)

Why does matrix multiplication exist for a linear transformation?

consider $T=(x_2, x_1)$, $T: \mathbb{R}^2 \to \mathbb{R}^2$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$T=(x_1-x_2, x_1+x_2)$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1-x_2 \\ x_1+x_2 \end{bmatrix}$$

$\hookrightarrow x_1\begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_2\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Ax ✓ either rotates/scales (changes the basis)

Basis is representative of vector space.

$Ax=b$ If A has LI vectors $\Rightarrow x=A^{-1}b$

$x=Bb$

$$T = Av = \begin{pmatrix} | & | \\ a_1 & a_2 \\ | & | \end{pmatrix} v = \begin{pmatrix} a_1 & a_2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} | & | \\ v_1 a_1 & v_2 a_2 \\ | & | \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$v=A^{-1}b$

T (on basis of $\mathbb{R}^n$)

$$T(\{b_1, \cdots, b_n\}) = \begin{pmatrix} | & | & & | \\ T(b_1) & T(b_2) & \cdots & T(b_n) \\ | & | & & | \end{pmatrix} = A$$

$A_{m\times n}\, v_{n\times 1} = x_{m\times 1} \in \mathbb{R}^m$

x: some lin comb of basis of corresp vector space

$$\begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$

$a_i \in \mathbb{R}^m$

$[A] \xrightarrow[\text{echelon}]{\text{row}} [A']$

pivot + free

↳ vector space of x will have basis with the pivot vectors

$V_{n\times 1} \xrightarrow{A} x_{m\times 1}$

vector space changes $\Rightarrow$ basis changes.

Q. $A_{6\times5}$ is a non-zero matrix. $b \in \mathbb{R}^6$. Then $Ax=b$ has

a) a soln for every b

b) no soln for some b

c) soln only when $b=0$

d) a soln b iff col^{m}s of A are LI

$$\rightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 5 & 6 & 4 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & -2 & -1 \\ 0 & -4 & -1 \end{array}\right]$$ Inconsistent opt. b

opt-d will be true if b is in col^m space of A. The 'iff' part makes the opn wrong.

Properties of linear transformations

$T(v) = Av$

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m \Rightarrow A_{m\times n}$

For T to be linear:

a) additivity

b) homogenity

Every linear transformn can be represented by a matrix A.

$T(0) = 0$.

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix} = T\left(\begin{bmatrix} | & & | \\ e_1 & e_2 \cdots & e_n \\ | & & | \end{bmatrix}\right)$$

$$= \begin{bmatrix} | & & | \\ T(e_1) & \cdots & T(e_n) \\ | & & | \end{bmatrix}$$

$T(v) = A(v)$

↳ a transformn can be seen as a funcn.

If the linear transformn is

a) one-to-one mapping → $f: X \rightarrow Y$ for every x there'll be exactly 1 image in y. $x \in X$, $y \in Y$.

b) onto mapping → for every $y \in Y$, there'll be atleast 1 pre-image $x \in X$.

many-to-one:

$T: \mathbb{R}^n \rightarrow \mathbb{R}^n \qquad T(v) = A_{n\times n} \cdot v$

If T is many-to-one,

$Av = b$

$b \in$ codomain $\qquad v \in$ domain

For a given matrix $A_{n\times n}$, it'll map every vector in domain to exactly 1 vector in codomain.

Every linear transformation is many-to-one.

onto

If T is onto,

codomain == range

For every b there's some pre-image v.

$v = A^{-1}b$

Then A is invertible. (If A is onto (and one-to-one))
↳ implicit

many to one



not onto

Null space exists.

A^{-1} doesn't exist.

one to one



onto as well

$Av = b$

Null space doesn't exist.

A^{-1} exists.



not one-to-one

not linear combination

onto

2 possibilities

① col^m space of A is basis for codomain.

② Basis for codomain is subset of col^m space.

] In either case, A is invertible.

---

many to one for $T: \mathbb{R}^n \to \mathbb{R}^m$ $\quad A_{m\times n}\, x = b$

① $n < m$ $(A_{3\times 2})$

range ≠ codomain

some b exists for which T is not valid from $\mathbb{R}^n$.

= some b exists for which there's no soln.

② $m < n$ $(A_{2\times 3})$

there'll be soln if rank(A) = n.

soln always exists → col^m space of A has dim = that of codomain

$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}_{2\times 3} x = b$ → soln may not exist for all b.

one-to-one for $T: \mathbb{R}^n \to \mathbb{R}^m$

1. $n < m$ $(A_{3\times 2})$

1. col^n of A are LI

one-to-one will hold

2. col^n of A are LD

one-to-one will not hold

3. If dim of col sp of A = dim (codomain)

⇒ not one-to-one.

2. $m < n$ $(A_{2\times 3})$

1. col^n of A can never be LI ⇒ no one-to-one.

Summary of previous stuff:

onto:

for every y in codomain, we have x in domain.

$T: \mathbb{R}^n \to \mathbb{R}^n$

If T is onto → every b will have a soln in $\mathbb{R}^n$

vectors in domain span the co-domain space

↳ so any transformn A will have n LI vectors.

col space(A) = basis of $\mathbb{R}^n$.

A is invertible

many to one:



we have some valid null space

A^{-1} doesn't exist

One to one



Q. check if matrix is one-to-one and onto:

(a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ onto ✗ 1-2-1 ✓

→ no pivot in 3rd row.
not onto

3×2
$T: \mathbb{R}^2 \to \mathbb{R}^3$
$m < n$

2 LI cols. can't cover $\mathbb{R}^3$.
not onto.

one-to-one: col vectors are LI.
yes. one-to-one.

(b) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$ onto ✗ 1-2-1 ✓

→

(c) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ onto ✓ 1-2-1 ✓

→

(d) $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

→ 1-2-1 ✗
(cols not LI)

onto ✗
(can't cover $\mathbb{R}^3$)

$T: \mathbb{R}^3 \to \mathbb{R}^3$

(e) $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

→

onto ✓
1-2-1 ✓

(f) $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}_{2\times 3}$

→

2×3
$\mathbb{R}^3 \to \mathbb{R}^2$
can cover $\mathbb{R}^2$.
onto ✓
1-2-1 ✗ (not LI)

Q. Let $T: \mathbb{R}^2 \to \mathbb{R}^3$. For 3 vectors in $\mathbb{R}^2$, prove that $\{T(v_1), T(v_2), T(v_3)\}$ is LD in $\mathbb{R}^3$.

→ $v_1, v_2, v_3 \in \mathbb{R}^2$

$T(v_1), T(v_2), T(v_3) \in \mathbb{R}^3$. To prove: these 3 are LD.

$A = \begin{bmatrix} | & | \\ A_1 & A_2 \\ | & | \end{bmatrix}$

$v_1 = \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow Av_1 = [aA_1 + bA_2] = T(v_1)$

$v_2 = \begin{pmatrix} c \\ d \end{pmatrix} \Rightarrow Av_2 = [cA_1 + dA_2] = T(v_2)$

$v_3 = \begin{pmatrix} p \\ q \end{pmatrix} \Rightarrow Av_3 = [pA_1 + qA_2] = T(v_3)$

Q. Consider $A_{m\times 1}$. $(AA^T)_{n\times n}$ will be a rank 1 matrix.

→ proof:

$A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$ $\quad AA^T = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} (a_1\ a_2 \cdots a_n) = \begin{bmatrix} | & | & & | \\ a_1 C & a_2 C & \cdots & a_n C \\ | & | & & | \end{bmatrix}$

C ← Rank 1 matrix.

Hy for $A_{m\times n}$, $(AA^T)_{m\times m}$ will have rank k for k lin. ind. cols in A. $(k < n)$

$A_{3\times 2}$ is 3×2.
A will have max rank 2.
$\therefore A[v_1\ v_2\ v_3]$ will have max rank 2

$A_{3\times 2}\begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}_{2\times 3}$

$\begin{bmatrix} | & | & | \\ T(v_1) & T(v_2) & T(v_3) \\ | & | & | \end{bmatrix}$

If we transform a LD set using any linear transformation, the resulting set will always be LD.

$T: \mathbb{R}^n \to \mathbb{R}^m$, $n < m$

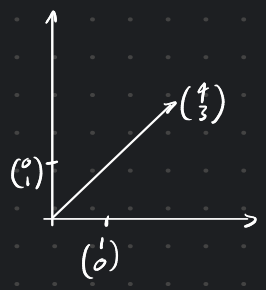
Q. $T: \mathbb{R}^3 \to \mathbb{R}^3$. $T(x,y,z) = 2x, y+z, x+y+z$. what is the rank of T?

→ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x \\ y+z \\ x+y+z \end{bmatrix}$

A

$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ rank = 2

Change of basis using transformation



$\begin{pmatrix}4\\3\end{pmatrix} = 4\begin{pmatrix}1\\0\end{pmatrix} + 3\begin{pmatrix}0\\1\end{pmatrix}$

$\left\{\begin{bmatrix}2\\1/2\end{bmatrix}, \begin{bmatrix}1/2\\2\end{bmatrix}\right\}$ is also a basis.

How do we represent $\begin{pmatrix}4\\3\end{pmatrix}$ using this basis?

$$\begin{bmatrix}4\\3\end{bmatrix} = c_1\begin{bmatrix}2\\1/2\end{bmatrix} + c_2\begin{bmatrix}1/2\\2\end{bmatrix}$$

$$\begin{bmatrix}4\\3\end{bmatrix} = \begin{bmatrix}2 & 1/2\\1/2 & 2\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix}$$

Since cols are independent, this is invertible.

$Ax = b$

$\Rightarrow x = A^{-1}b$

$$A|I \rightarrow \left[\begin{array}{cc|cc}2 & 1/2 & 1 & 0\\1/2 & 2 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & 1/4 & 1/2 & 0\\1 & 4 & 0 & 2\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & 1/4 & 1/2 & 0\\0 & 15/4 & -1/2 & 2\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & 1/4 & 1/2 & 0\\0 & 1 & -2/15 & 8/15\end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc}1 & 0 & 16/30 & -2/15\\0 & 1 & -2/15 & 8/15\end{array}\right]$$

$\frac{1}{2} + \frac{1}{2}$ $\qquad C_{A^{-1}}$

$$A^{-1}b = \begin{bmatrix}8/15 & -2/15\\-2/15 & 8/15\end{bmatrix}\begin{bmatrix}4\\3\end{bmatrix}$$

$$\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}32/15 - 6/15\\-8/15 + 32/15\end{bmatrix} = \begin{bmatrix}26/15\\24/15\end{bmatrix}$$

$b = Ax$ — standard basis

$x = A^{-1}b$ — in new basis (cols of A)

$Ax = b$ — A transforms x to b in basis of A.

$\begin{bmatrix}4\\3\end{bmatrix}$ in basis of $\left\{\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}0\\2\end{bmatrix}\right\}$

Inverse of $\begin{bmatrix}1 & 0\\-1 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\1/2 & 1/2\end{bmatrix}$

$$\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}1 & 0\\1/2 & 1/2\end{bmatrix}\begin{bmatrix}4\\3\end{bmatrix} = \begin{bmatrix}4\\7/2\end{bmatrix}$$

A^{-1} b

$\{a_1, a_2\}$ is the basis for space B.

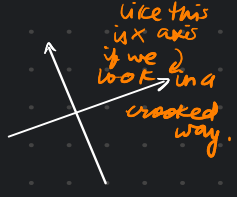
$A = \begin{bmatrix}a_1 & a_2\\1 & 1\end{bmatrix}$

a_1 in basis B is given as:

$$\begin{bmatrix}a_1 & a_2\\1 & 1\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix} = a_1$$ (std basis)

$$\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}a_1 & a_2\\1 & 1\end{bmatrix}^{-1}a_1 = \begin{bmatrix}1\\0\end{bmatrix}$$

Any basis vectors will look like std basis vectors for that basis itself.



vector in basis B_1 in another basis B_2 :

$v_1 \leftarrow B_1$ corr. matrix is B_1 for basis B_1

$v_2 \leftarrow B_2$

$$v_2 = ? \; v_1$$

v_1 is in B_1 basis. change to std basis.

$B_1 v_1$

now $B_1 v_1$ is in std. basis. change to B_2.

$B_2^{-1} B_1 v$

$$\Big\} \; v_2 = B_2^{-1} B_1 v_1$$

Orthogonal spaces

- If V is orthogonal to W, $V \in \mathbb{R}^n$
  then ① $W \in \mathbb{R}^n$
  ② all vectors in W are $\perp^r$ to space V and vice versa.

Q. given 2 subspaces, can we say if they are orthogonal?
→ ① They have to be subspaces of the same original space, say $\mathbb{R}^n$.
② Their rank should add upto n.



Using 4 fundamental subspaces:

$R(A) \perp N(A)$ $\quad$ $C(A^T) \perp N(A)$

$R(A^T) \perp N(A^T)$ $\quad$ $C(A) \perp N(A^T)$

Q. Given some subspace V, which of the foll^n subspaces can be considered as orthogonal subspaces to V?

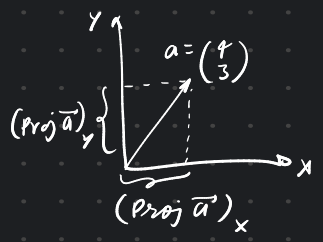
↳ E.g. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$. V is col^m space of A.

Possible orthogonal subspaces:

① Null space of A^T = $N(A^T)$ ← find using $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

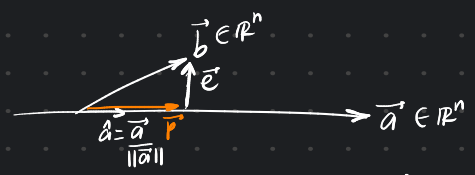
My check others.

Projections



projectn of $\vec{a}$ on x-axis = $(\text{proj}\ \vec{a})_{x\text{-axis}} = 4$

Projection onto a line :



$\vec{p}$ = some scalar multiple of $\vec{a}$

$\vec{b} = \vec{p} + \vec{e}$

$A\vec{x} = \vec{a}$ if b is NOT in basis A, we can't solve $Ax = b$

But we could approximate,

$\vec{a}\hat{x} = \vec{p}$ $\quad \vec{p} = \vec{a}\hat{x}$

$b = p + e$

$\Rightarrow \vec{e} = \vec{b} - \vec{p}$

$\vec{e} \perp \vec{p} \Rightarrow \vec{p} \cdot \vec{e} = 0$

$\Rightarrow p^T e = 0$

$\vec{a} \perp \vec{e}$

$\Rightarrow a^T e = 0$

$a^T(b - a\hat{x}) = 0$

$\Rightarrow a^T b = a^T a \hat{x}$

$$\hat{x} = \frac{a^T b}{a^T a}$$

HW1 A is a sq. matrix & has LI columns then A^TA & AA^T both are invertible. Prove this.

→ This goes for rectangular matrices as well.

→ $A_{n\times n}$ has LI cols. A^TA and AA^T will be $n\times n$.

→ Basis will have n vectors.

A has n LI cols ⇒ n LI rows

⇒ A^T has n LI cols ⇒ n LI rows.

$$AO = A\begin{pmatrix} | & | & & | \\ b_1 & b_2 & \cdots & b_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ Ab_1 & Ab_2 & \cdots & Ab_n \\ | & | & & | \end{pmatrix}$$

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}$$

If b_1, b_2 are LI, then Ab_1 and Ab_2 will be LI. ⇒ $v_1, v_2, \cdots, v_n$ are LI.

$b_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ $b_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ $b_1 \neq c \cdot b_2$

$Ab_1 \neq Ac\,b_2$ ⇒ $Ab_1 \neq c \cdot Ab_2$ ⇒ LI

pair(a c) ≠ pair(b d)

(a c) ∦ (b d)

A has n LI cols.

$$A^TA = A^T\underbrace{\begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix}}_{\text{LI cols.}} = \underbrace{\begin{bmatrix} | & | & & | \\ A^Ta_1 & A^Ta_2 & \cdots & A^Ta_n \\ | & | & & | \end{bmatrix}}_{\text{LI cols}}$$

$$A = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}_{3\times 3} \qquad B = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 3 \\ 1 & 1 & 5 \end{pmatrix}$$

A is LD — LI cols

$v_1 = v_2$

$$AB = \begin{pmatrix} | & | & | \\ v_1-v_2+v_3 & -v_1+v_2+v_3 & * \\ | & | & | \end{pmatrix} = \begin{pmatrix} | & | & | \\ v_3 & v_3 & * \\ | & | & | \end{pmatrix}$$

LD

→ So cols of A should be LD as well for resulting matrix to have LI cols as well.

In $AB=C$, for C to have LI cols, A and B should have LI cols.

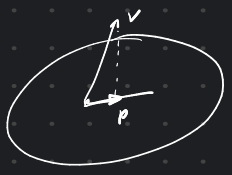
- If $A_{m\times n}$ has only LI cols, then AA^T and A^TA are invertible. (proof for rectangular matrices later)

For matrix, if $T: \mathbb{R}^m \to \mathbb{R}^n$ is not invertible, then it will never have range = codomain.

→ False for rect. matrices

→ True for sq. transforma"s.] → can't cover all of $\mathbb{R}^n$ space as col^s of A are not L.I.

Projection of a vector onto subspace B



$Ax = v$ ← If we can't solve this,
$A\hat{x} = p$ ← we solve for an approximate soln

subspace $B \in \mathbb{R}^n$
↓
$\{a_1, a_2, \dots, a_n\}$ are LI basis vectors.

$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix}$ we cannot solve for $Ax = v$ ✓ v is not in subspace of A.

we need to approximate to $A\hat{x} = p$

$A^T e = 0 \Rightarrow A^T(b-p) = 0$
$\Rightarrow A^T(b - A\hat{x}) = 0$
$\Rightarrow A^T b - A^T A\hat{x} = 0$
$\Rightarrow A^T b = A^T A\hat{x}$
$\hat{x} = (A^T A)^{-1}(A^T b)$

Composition of linear compositions

$(T_1 \circ T_2)(v) = T_1(T_2(v))$
→ we need to check which transformation is valid to check which one to apply first.

Lin. transformation wrt some basis B

$v|_B \rightarrow v'|_e$

$v' = B\,v|_B$ → change of basis

$T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ defined as $Av = T(v)$

$T(v) = Av' = AB v$

$T(v) = AB v|_B$ in some other basis B
↳ In std basis

$S|_B = B^{-1} A B v = Mv$ where $M = B^{-1} A B$
↳ In basis B

Input in B_1, output B_2: ✓ How to reduce computation:

$S|_{B_2} = B_2^{-1} A B_1 v|_{B_1}$

T: — —

$A_{5\times 3}$ $A_{n\times n}$

which space the range of A belongs to?

a) R^4 b) R^3 c) R^5 d) R^6

·) Range ∈ col^n space

·) Domain ∈ row space.



8. A^T is invertible iff A is invertible. T/F?

→ $A^{-1}A = I$

$(A^{-1}A)^T = I^T = I$

$A^T(A^{-1})^T = I$

A is invertible → n LI cols
n LI rows

↓

A^T will have n LI cols & n LI rows.

Permutation matrix

permutation matrix P has rows only from the identity matrix in some order.

e.g. $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

- $PP^T = I \Rightarrow P^{-1} = P^T$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$